

Walmart Sales Analysis Report

Asking the Right Questions Series: Day 1

DATA OVERVIEW

This report focuses on historical sales data for Walmart stores, spanning the period from **February 5, 2010** to **November 1, 2012**. The dataset encompasses various economic and environmental factors influencing weekly performance.

Dataset Source: kaggle.com/datasets/yasserh/walmart-dataset

Primary File: Walmart_Store_sales

Feature Definitions

Store	The unique identifier for each store.
Date	The specific week of sales.
Weekly_Sales	Total sales recorded for the given store in that week.
Holiday_Flag	Binary indicator (1 for Holiday week, 0 for Non-holiday week).
Temperature	The temperature in the region on the day of sale.
Fuel_Price	The prevailing cost of fuel in the region.
CPI	The Consumer Price Index for the period.
Unemployment	The prevailing unemployment rate in the region.

ANALYSIS FRAMEWORK

Phase 1: Basic EDA & Integrity

- Identify missing values and determine the overall shape of the dataset.

- Analyze the data types assigned to each feature.
- Conduct a search for anomalies or outliers within the data.
- Confirm the specific years represented in the dataset.
- Identify the total count of unique stores.
- Calculate the average weekly sales across all stores.
- Determine the average, mode, and median temperature recorded.
- Identify the maximum, minimum, and mean fuel prices.
- Analyze the distribution of the unemployment rate and how it has changed year-over-year.
- Quantify the total count of holiday flags per year for each store.

Phase 2: Intermediate & Relational Analysis

- Evaluate the distribution of unemployment on a per-store basis.
- Calculate the yearly average sales generated by each store.
- Assess the relationship between temperature and fuel price.
- Analyze the correlation between Weekly Sales, Holiday Flag, CPI, and Unemployment (utilizing heatmaps for visualization).
- Break down the yearly fuel prices paid, distributed per store across different years.
- Identify which store produces the highest weekly sales volume.

- Evaluate store performance during holiday weeks compared to non-holiday weeks on a yearly basis.
-
- Investigate if temperature fluctuations demonstrate a significant impact on weekly sales.
-
- Examine the relationship between each feature and Weekly Sales to identify the strongest predictors.
-
- Determine which year yielded the lowest total weekly sales across the network.
-
- Establish whether a positive or negative correlation exists between weekly sales and the unemployment rate.
-